

India Industrials and Infrastructure

India Energy Security - #1 Coal Gasification: The color of energy doesn't matter anymore...



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In this report, we dive deeper into one of the [5 key themes](#) we highlighted due to the M.East crisis - Coal Gasification. On paper it looks the easiest answer to solve part of India's big gas import bill, but despite a long history of discussions it has never really taken-off. Now given the push of the war we see high chances of on ground progress with policy support on supply side (e.g. capital support announced [yesterday](#)), and possibly even on demand side (long term off-take agreements, blending requirements).

a) What is coal gasification and learning from global examples: Coal gasification turns solid coal into synthetic gas (syngas - Hydrogen and CO) by subjecting coal to high temperatures and pressures in the presence of steam. This syngas can be used further downstream to make urea, methanol etc. **South Africa-** Sasol (not covered) built the world's largest coal-to-chemicals complex in 1980 using high-ash coal given oil embargoes on the country. **China** with a similar resource mix as India has scaled coal gasification from almost nothing to ~340 MTPA through policy support, and today gets ~85% of its ammonia and ~75% of its methanol from coal.

b) Strategic imperative for India: India has ~401 Bn tonne of coal resources (one of the largest globally), and reserves not very far from China- but while China mines ~4.5 BnTPA, India does just 1 BnTPA. India has just ~2 MTPA of coal gasification capacity. Hence, with India's import dependence of 90% for methanol, 100% for ammonia, and 20% for Urea - maximizing coal gasification will drive-a) Energy security, b) Limits Forex outflow, c) Fertilizer subsidy of ~ Rs 1 Trn given by govt. stays within the country.

c) Govt. support: In 2020, govt announced a plan to take coal gasification to 100 MTPA. Subsequently, a) VGF of Rs 85 Bn was announced, b) Regulated coal pricing allocated, c), Guaranteed ROE for specific govt. plants. Yesterday govt. further announced Rs 375 Bn capital subsidy - effectively 15% of capex expected to be mobilized towards the target.

d) Commercials & Risks: Considering the announced capex of coal gasification plants in India (+ latest capex support), plant utilization of 80%, and forex at ~ 95 INR/USD, we get a break-even for making Urea via coal gasification at \$11.7/MMBtu of LNG JKM price, vs ~ \$17/MMBTU currently and 3-year avg. LNG price of \$12.6/MMBTU- refer [Exhibit 17](#), [Exhibit 19](#). **Key risks:** a) The biggest is the absence of a long term off-take agreement - given high upfront capex, b) High Ash Indian Coal- makes it challenging to gasify, on which BHEL(not covered) is working, c) Competition from green NH3, whose economics are getting better with time.

e) Key players ([Exhibit 23](#)): The only big operational plant is owned by Jindal Steel & Power and 8 projects worth ~₹880 Bn are underway. Coal India, GAIL and BHEL are involved in most projects as equity holders or as equipment/fuel/ infra providers. NTPC targets to produce 5-10 MTPA of syngas by 2029 and L&T won the downstream construction package for one of the plants. Thermax is in discussion with CIL & MoC, to be made an equipment provider for the planned projects. Please note we cover only L&T and NTPC.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	13 May 2026 Closing	Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
			Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
LT.IN (L&T)	O	INR	3,915.80	4,637.00	(32.6)%	INR	109.00	123.00	155.00	22.8	20.1	16.7
NTPC.IN (NTPC Ltd)	O	INR	390.45	430.00	(26.9)%	INR	24.70	25.64	27.02	15.8	15.2	14.5
POWF.IN (Power Finance)	O	INR	446.00	460.00	(32.2)%	INR	69.67	86.54	88.18	6.4	5.2	5.1
RECL.IN (REC)	O	INR	346.55	450.00	(53.2)%	INR	60.32	65.52	67.62	5.7	5.3	5.1
ASIAX			1,965.26									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

LT.IN valuation is EV/EBITDA (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

Within our coverage, we think L&T is the beneficiary of a push for coal gasification on equipment side. Other than L&T, PFC-REC could see a new lending opportunity as it fits with their long term infra lending focus. On asset owners, NTPC's plan to leverage its coal resources for gasification could get accelerated with latest capital support announcement.

DETAILS

1. COAL GASIFICATION & LEARNINGS FROM OTHER MARKETS:

WHAT IS COAL GASIFICATION?

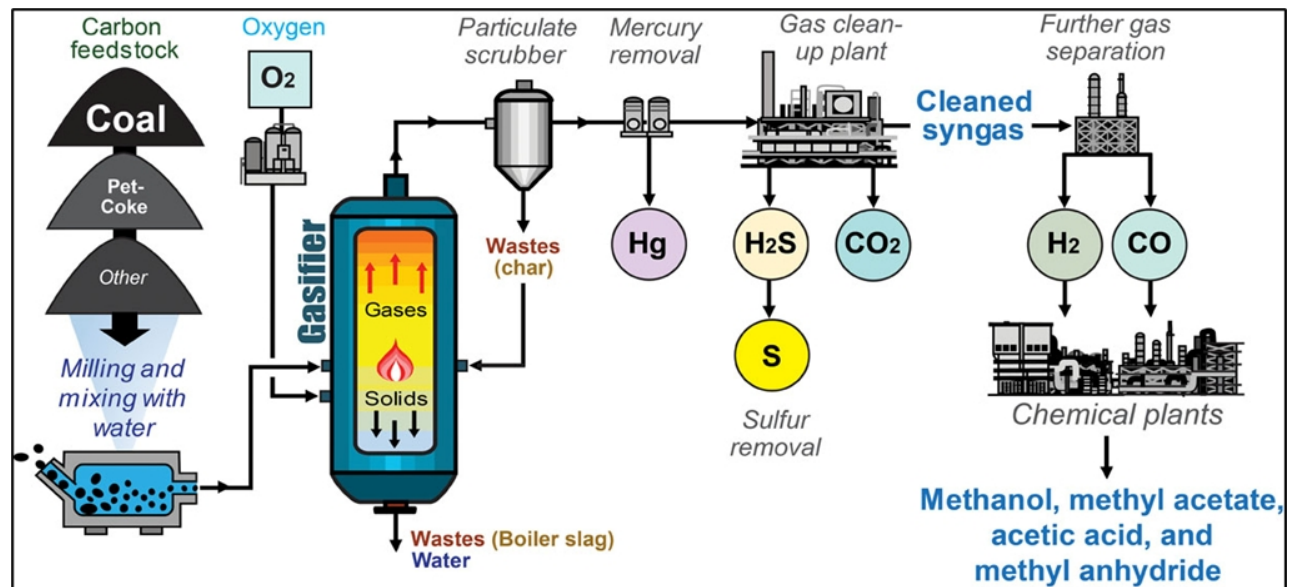
Gasification is a thermochemical process that subjects solid coal to high temperatures and pressures in the presence of steam and a controlled, limited amount of oxygen. Instead of simply burning the coal, this process breaks it down at a molecular level to produce synthesis gas or “syngas”- a versatile gaseous mixture consisting primarily of carbon monoxide and hydrogen. This syngas acts as a foundational chemical building block that can be directly synthesized into high-value industrial products, including ammonia, urea, methanol or used as a clean reducing agent in steel production.

A single coal feed produces methanol, acetic acid, hydrogen, and power simultaneously. Unlike an LNG terminal locked into natural gas, a gasification complex can pivot between products as commodity economics shift. NITI Aayog's Methanol Economy program— highlights that the syngas producing ammonia can produce methanol for potential petrol blending.

There are majorly 3 types of gasification technologies (or gasifiers):

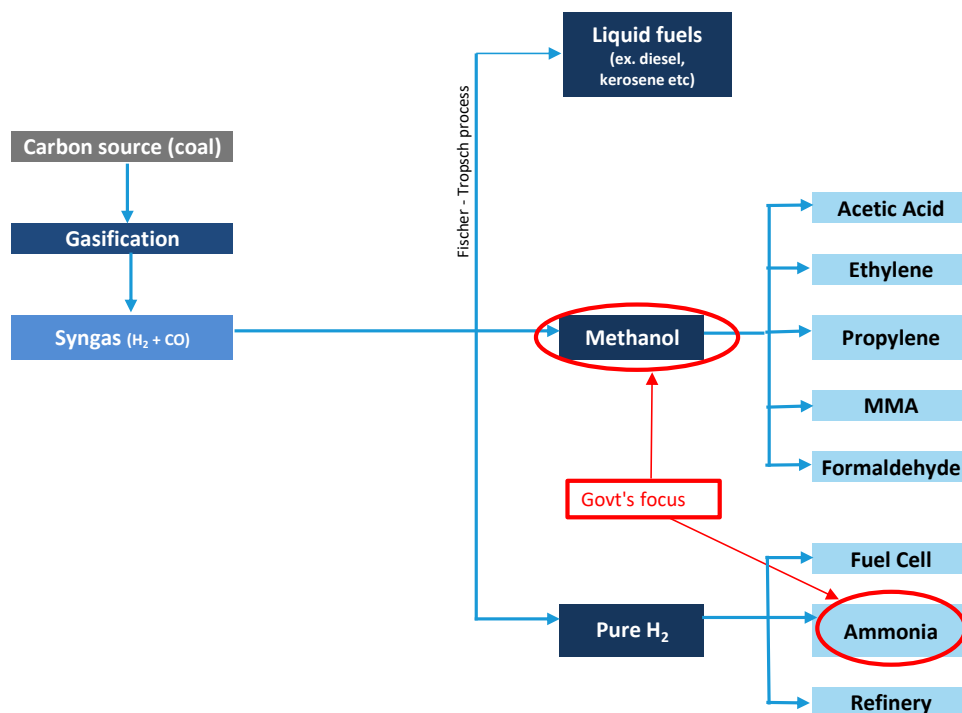
1. Entrained Flow Gasifier (Used by China)
2. Fixed Bed Gasifier (Used by South Africa)
3. Fluidized Bed Gasifier (**Most suitable for low-grade & high-ash coal found in India**) - BHEL (not covered) is currently developing this indigenously.

EXHIBIT 1: Coal Gasification Process



Source: University of Kentucky

EXHIBIT 2: Coal gasification - Downstream use cases

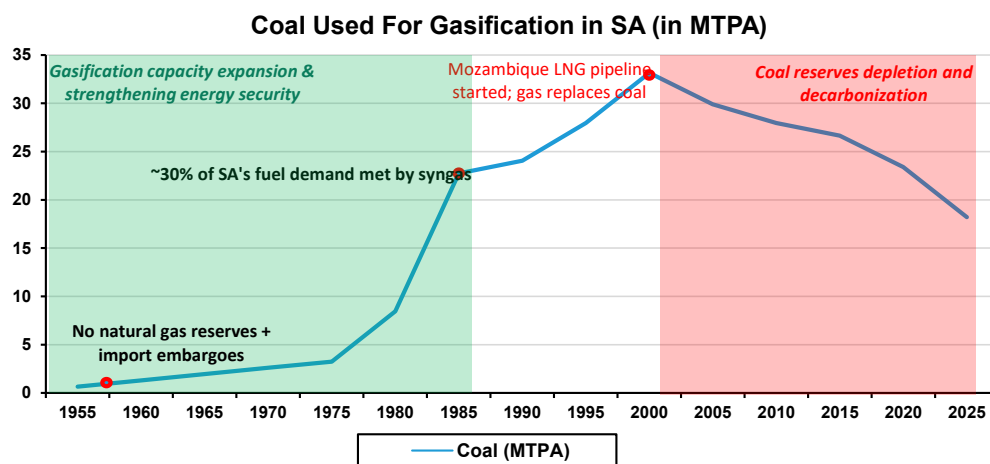


Source: CIL, Bernstein analysis

SOUTH AFRICA: SASOL – THE ORIGINAL COAL GASIFICATION CASE STUDY

Sasol's first gasification plant was built in the 1950s under apartheid oil sanctions. Sasol aggressively expanded during the '80s, was sustained by state ownership, a Strategic Fuel Fund levy on imported petroleum that subsidized domestic synfuels, and govt-guaranteed long-term off-take contracts insulating project economics from price volatility. When built, coal gasification was economically sub-optimal as crude oil was available at \$3–5/barrel. By the 1990s as oil price shot up to \$20–30/barrel, coal gasification became viable. Sasol now licenses its technology globally.

EXHIBIT 3: South Africa's coal consumption history for gasification



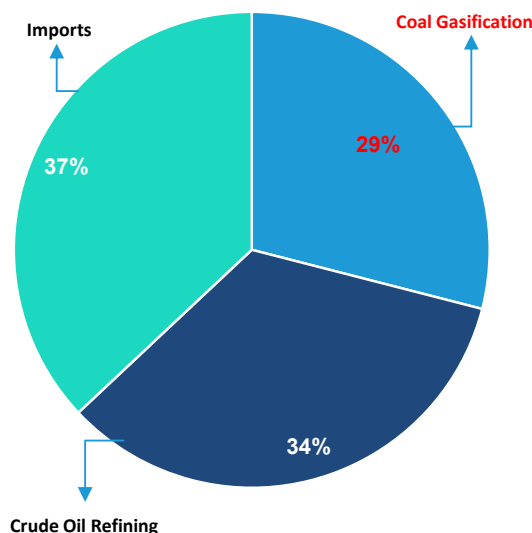
Source: Sasol reports, Bernstein analysis

South Africa strengthened its energy security by pioneering commercial scale coal gasification. By converting low-grade coal into syngas, the country produces methane-rich gas (MRG) and provides roughly 29% of its locally refined liquid fuel requirements (ex. diesel, kerosene etc.). The coal-to-liquids substitution saves the nation billions in forex annually and actively insulates its industrial economy from the extreme price volatility and supply shocks of global LNG and crude oil markets.

S. Africa has proven the use of high-ash coal, leading the way for India to build on it further (SA 25%, India 35%). JSPL has proven at Angul, Odisha. BHEL's indigenous BCGCL effort follows Sasol's playbook.

EXHIBIT 4: Liquid Fuel Production Mix in South Africa

South Africa's Liquid Fuel Supply, FY23

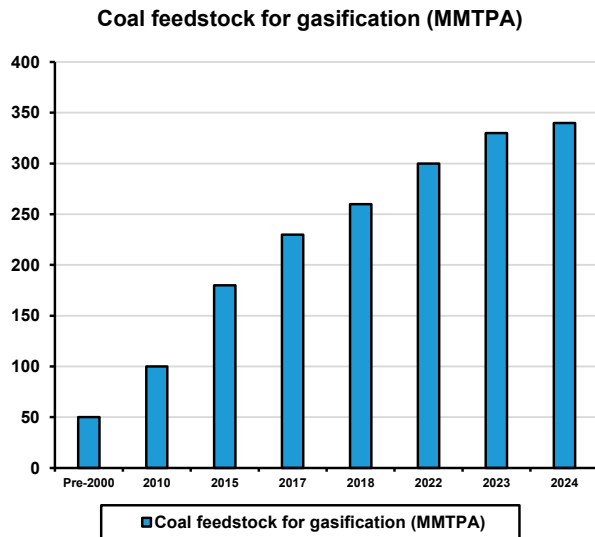


Source: NETL, Sasol reports, Bernstein analysis

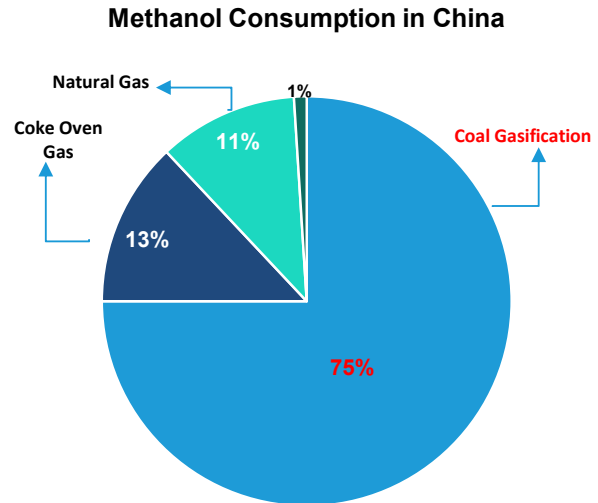
CHINA - REDUCING IMPORT DEPENDENCE

South Africa & China mastered coal gasification with the same playbook: sustained state interventions bridging the gap between high upfront capital costs and long-term import-substitution payoff. China's 2000s-2010s scale-up went further: state-owned enterprises (or SOEs) (Shenhua, Sinopec etc.) deployed coal-to-chemical plants using preferentially-priced captive coal allocations, state-bank debt at well below commercial rates, mandatory grid off-take of coal-derived synthetic natural gas, and tolerance for years of accounting losses absorbed by SOE balance sheets.

Today, about 78% of China's urea output comes from coal rather than natural gas. This coal gasification production model gives China access to abundant, domestically sourced energy, reducing exposure to volatile international gas prices and supply disruptions. Due to the Iran war, which has disrupted shipping through the Strait of Hormuz, the urea prices outside China surged by about 70%. In contrast, China maintained ample stocks, keeping domestic prices roughly a third of international benchmarks, all thanks to its large-scale coal-based production of urea. Today, China is among the largest exporters of fertilizers - shipping more than \$13 billion worth last year.

EXHIBIT 5: China: Coal gasification evolution history (approx. figures)

Source: CREA, IER, Bernstein analysis

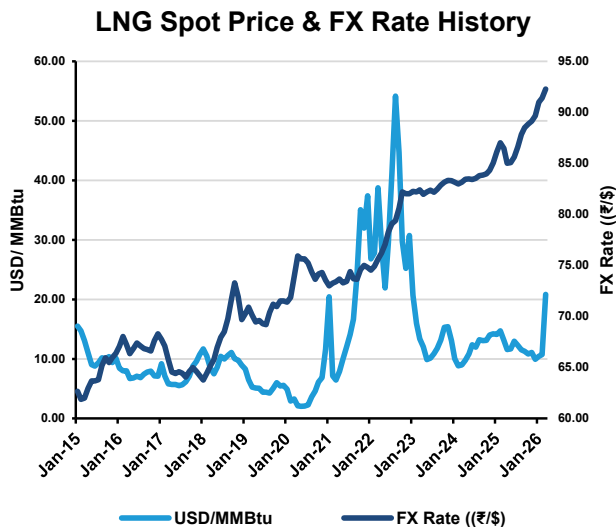
EXHIBIT 6: Split of Methanol Consumption in China, FY24

Source: US EIA, Bernstein analysis

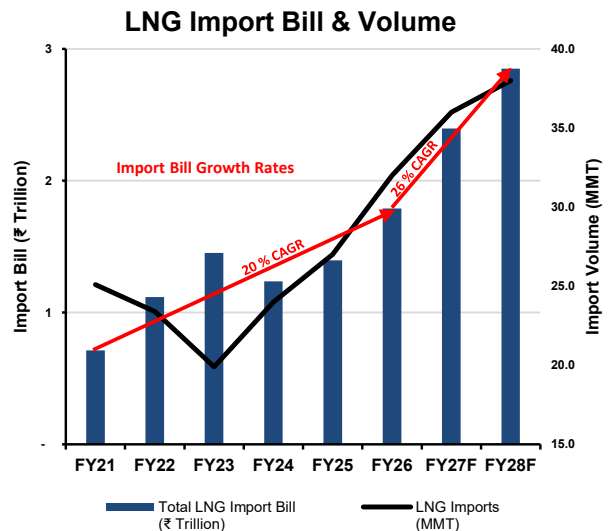
2. COAL GASIFICATION IN INDIA

WHY DOES INDIA NEED COAL GASIFICATION - THE BIG LNG BILL AND FERTILIZER SUBSIDIES...

The IEA projects India's total LNG import bill to grow at a 26.2% CAGR to ₹2.85 Trillion by FY2028F ([Exhibit 8](#)), driven by three forces: (i) absolute volume growth as industrial gas demand expands with GDP, (ii) structural FX depreciation at ~3.3%/yr. [Exhibit 7](#) illustrates the combined impact of LNG price volatility and FX depreciation on India's annual import bill. India's LNG bill in FY2023 hit ₹1.45 Trillion.

EXHIBIT 7: India LNG import bill compounded by commodity price x Forex

Source: Platts JKM, RBI, Bernstein Analysis

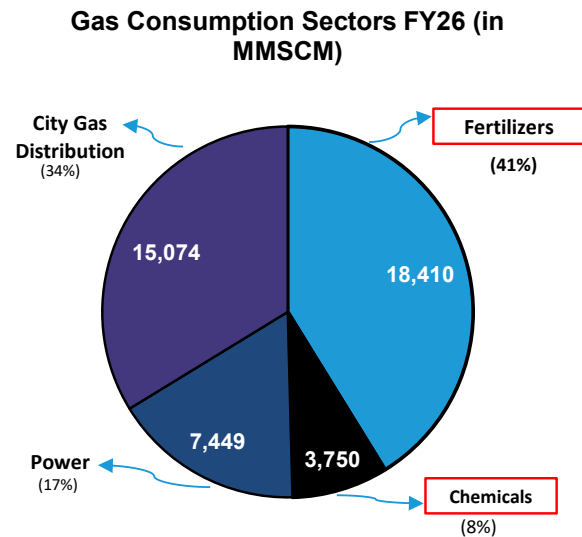
EXHIBIT 8: IEA forecasts this import bill to continue to rise sharply

Source: PPAC, IEA, Bernstein Analysis

Fertilizer support bill: Exhibit 9 shows India's gas consumption by sector from FY2021–22 through FY2025–26. Fertilizers consume ~40% of India's imported gas every year. It is the government's highest-priority target for coal gasification substitution.

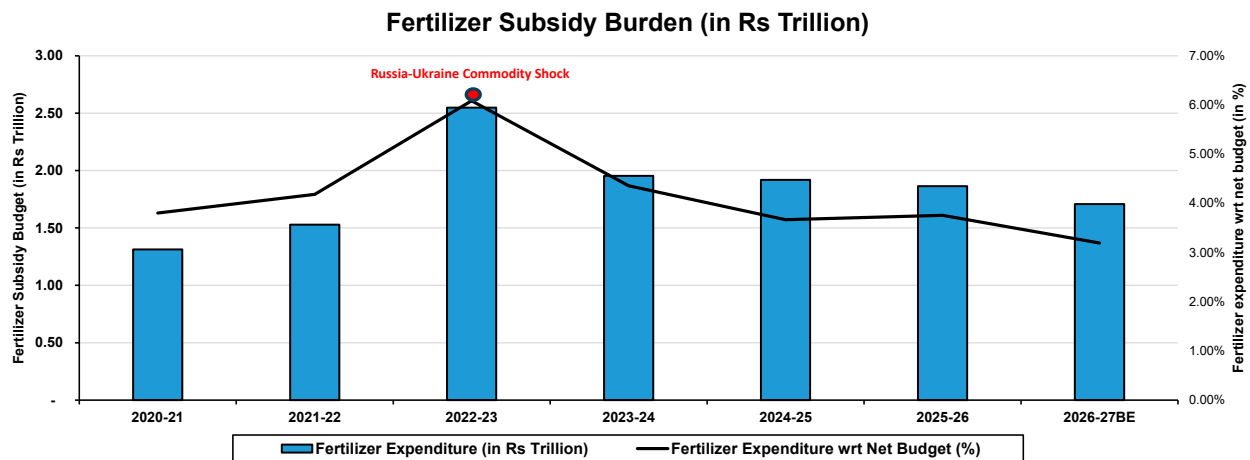
The fertilizer subsidy burden (Exhibit 10) for the Indian govt peaked at ₹2.5 trillion in FY2023 — ~6% of the entire net budget — created by the LNG price spike due to the Russia-Ukraine war along with Rupee depreciation in FY23. Coal gasification could convert this open-ended, LNG-contingent liability into a fixed, rupee-denominated cost — independent of global gas prices.

EXHIBIT 9: **40% of gas usage in India is in Fertilizer industry**



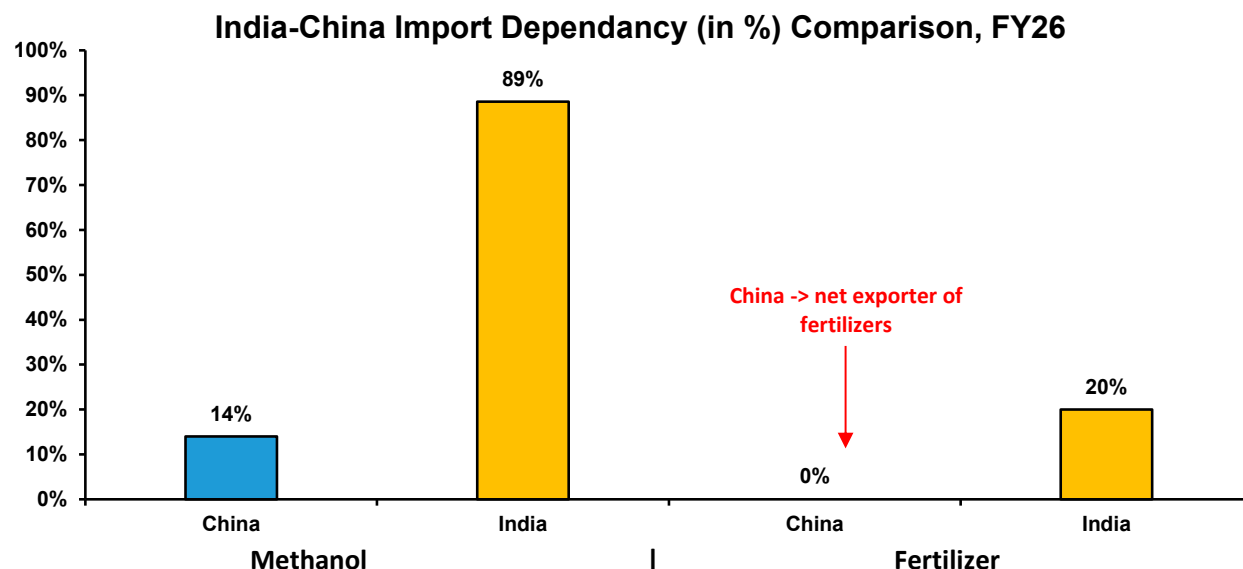
Source: Niti Aayog, Bernstein Analysis

EXHIBIT 10: **There is a big fertilizer subsidy burden on the govt. which is primarily supporting LNG / Ammonia imports**



Source: India Budget report, Sansad report, PIB, Bernstein analysis

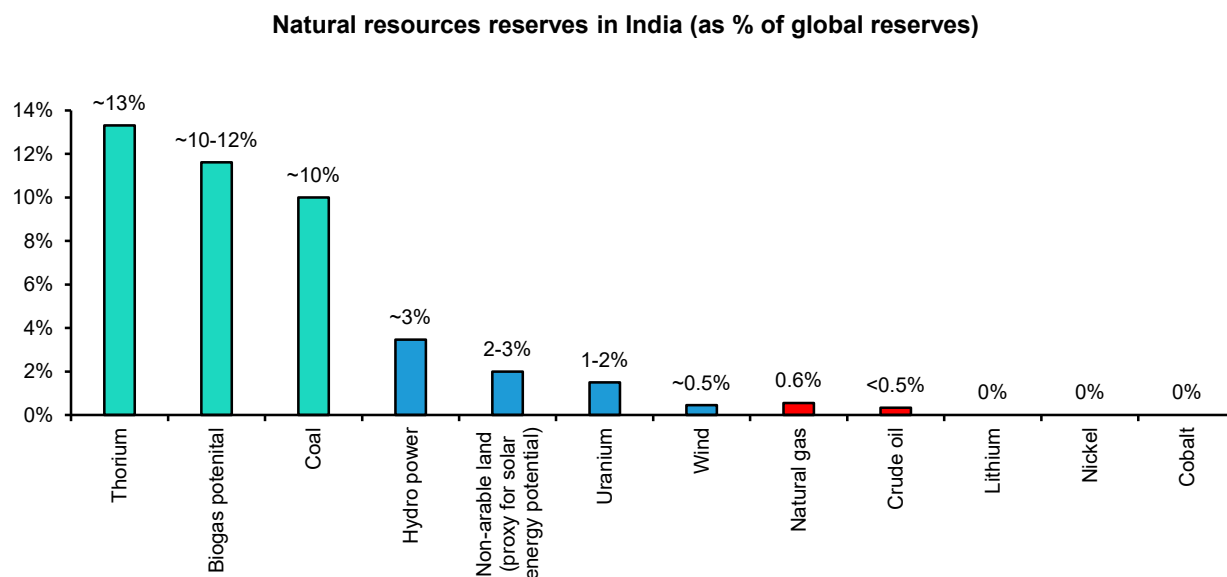
China despite having low oil-gas reserves has successfully reduced import dependence for Methanol and fertilizers with support of coal gasification. India, with a similar resource mix, has not capitalized on its potential yet.

EXHIBIT 11: China has managed to reduce this import dependency with coal gasification, now India needs to step up

Source: Ministry of Chemicals & Fertilizers, Bernstein analysis

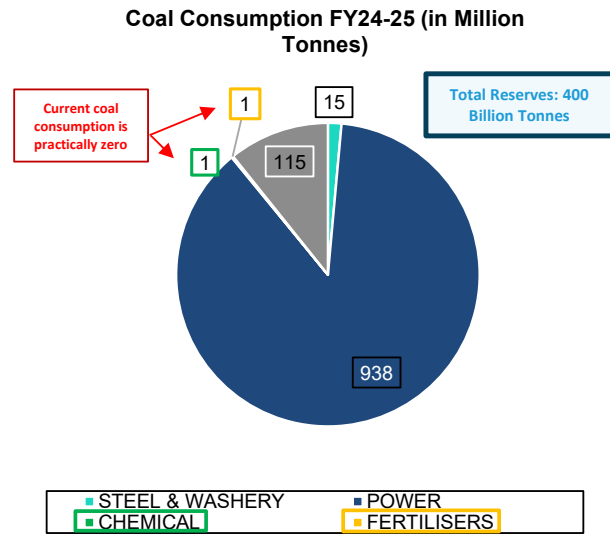
UNTAPPED COAL RESOURCES

India's structural gap between its coal reserves & utilization: India has 400 billion tonnes of coal reserves — the fifth-largest in the world — yet currently has coal gasification capacity of just ~ 2.4 million tonnes per year, almost entirely at JSPL's Angul plant.

EXHIBIT 12: India share of global natural resources - Needs to make the most of its large reserves

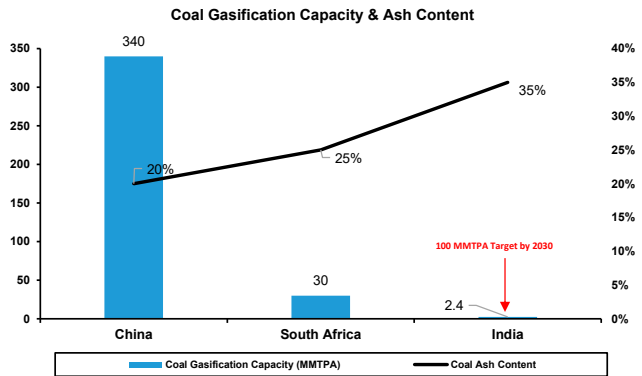
Source: Govt. data, press articles, Bernstein analysis

EXHIBIT 13: Coal consuming sectors in India



Source: NITI Aayog, Bernstein analysis

EXHIBIT 14: China-SA-India Comparison in Coal Gasification



Source: CREA, Bernstein analysis

3. GOVT. POLICY TAILWINDS & PLANNED GASIFICATION PROJECTS

THE POLICY ARCHITECTURE

India's coal gasification mission was formalized in 2020 under the Atmanirbhar Bharat reforms, with the Ministry of Coal setting a target of 100 MT of coal to be gasified annually by 2030. For the next three years, the mission produced ministerial announcements, project MoUs, and revived JV structures (TFL, BCGCL) - but no bankable financial framework, leaving every project sub-economic against imported LNG. That gap closed in 2024, when three critical events arrived within a 12-month window:

- January 2024 – VGF Scheme (₹8,500 Cr):** CCEA approved viability gap funding across three categories. Cat I (PSU JVs): ₹1,350 Cr per project (3 approved). Cat II (private): ₹1,000 Cr per project (3 approved). Cat III (demo): ₹100 Cr (1 approved). Directly bridges the CAPEX premium of the coal route vs LNG route.
- September 2024 – DoF 12% IRR Guarantee:** Department of Fertilizers confirmed guaranteed 12% post-tax equity IRR for **Talcher Fertilizers Ltd. (TFL)** investors for the first 8 years of production. This converts TFL equity into a quasi-sovereign return floor – removing project economics as an investment barrier.
- December 2024 – ROM Regulated Price for Gasification:** Ministry of Coal directed CIL to supply coal to gasification projects at notified regulated price (₹3,000–3,500/MT) vs NRS auction price (₹4,500–5,500/MT). A

EXHIBIT 15: Govt. incentives to promote coal gasification

S. No.	Date	Incentive	Details
1	24-Jan-24	VGF / Capital Grant — Category I (PSUs)	₹1,350 Cr per project or 15% of capex, whichever is lower; up to 3 PSU projects; total pool ₹4,050 Cr
2	24-Jan-24	VGF / Capital Grant — Category II (PSU + Private)	₹1,000 Cr per project or 15% of capex, whichever is lower; competitive bidding; pool ₹3,850 Cr
3	24 Jan 24	VGF / Capital Grant — Category III (Demo / Small-Scale)	₹100 Cr per project or 15% of capex; min capex ₹100 Cr; min syngas output 1,500 Nm ³ /hr; pool ₹600 Cr
4	2022 (reaffirmed 2024)	Revenue Share Rebate — Gasification Coal	50% rebate on revenue share in commercial coal block auctions, provided ≥10% of mine's production is used for gasification
5	Dec-24	Regulated / Floor Coal Price	Coal supply to gasification plants at CIL ROM regulated price (₹3,000–3,500/MT); floor price at notified regulated-sector price for projects commissioning within 7 yrs
6	Oct–Nov 2025	UCG-Specific Incentives	Floor revenue share reduced to 2%; upfront mine auction amount fully waived; 50% rebate on performance security
7	24-Jan-24	CIL Equity Participation (beyond 30% limit)	Cabinet approved CIL to breach statutory 30% equity ceiling to form JVs with BHEL (Lakhanpur) and GAIL (Sonepur Bazari), de-risking project finance
8	Sep-24	DoF IRR Guarantee (Fertilizer JVs)	12% post-tax equity IRR guaranteed for 8 years from production date for eligible PSU JV fertilizer gasification projects (e.g. TFL)

Source: Ministry of Coal, Bernstein analysis

30–40% feedstock cost reduction that improved total project economics by 6–8 percentage points.

On May 13th 2026, the govt. approved a 'Scheme for Promotion of Surface Coal/Lignite Gasification Projects' with a financial outlay of Rs.37,500 Crore. The Government has also extended coal linkage tenure up to 30 years (initially it was 15yrs), providing long-term policy certainty for investment in coal gasification projects. This long-term certainty de-risks the project cash flows as well as aligns the coal contract duration with the actual useful life of the plant and machinery.

THE PLANT UNIVERSE – ₹880B IN ANNOUNCED PROJECTS

[Exhibit 16](#) presents the full India gasification plant universe as of April 2026. Nine projects totaling ~₹880B in announced CAPEX, with government financial support confirmed for all VGF-eligible projects. The four PSU-backed mega-projects — **TFL**, **BCGCL**, **CGIL**, **CIL-BPCL JV** — constitute the investable core at ~₹690Bn combined. JSPL's operating plant at Angul is the only fully commissioned reference.

EXHIBIT 16: **Under-construction and planned coal-gasification capacity**

Project / Entity	Players	Estimated Capex	Planned Capacity	Govt Financial Support
Under construction projects				
Jindal Steel & Power (JSPL) Expansion	Jindal Steel	~₹38B	2 MTPA Coal Gasification for DRI (+2.4 MMTPA already operational)	₹569.05 Cr lump sum grant
Talcher Fertilizers Limited (TFL)	GAIL, CIL, RCF, FCIL	~₹190B	1.27 MMTPA 'Neem' Coated Urea	Guaranteed 12% post-tax equity IRR policy
Bharat Coal Gasification & Chemicals Ltd (BCGCL)	CIL, BHEL	~₹250B	0.66 MMTPA Ammonium Nitrate	₹1,350 Cr lump sum grant
New Era Cleantech Solution (Integrated Complex)	New Era Cleantech Sol.	~₹70B	0.33 MMTPA Ammonium Nitrate & 0.1 MMTPA Hydrogen	₹1,000 Cr lump sum grant
Planned				
Greta Energy & Metal Pvt Ltd	Greta Energy	~₹28B	0.5 MTPA Direct Reduced Iron (DRI)	₹414.01 Cr lump sum grant
Coal Gas India Ltd (CGIL)	CIL, GAIL	~₹130B	0.5 MMTPA Synthetic Natural Gas (SNG)	₹1,350 Cr lump sum grant
CIL-BPCL Joint Venture	CIL, BPCL	~₹122B	0.5 MMTPA Synthetic Natural Gas (SNG)	₹1,350 Cr lump sum grant
NLC India Limited (Lignite-to-Methanol)	NLC	~₹44B	0.40 MMTPA Methanol	Seeking PLI scheme inclusion

None of them is covered by us
Source: Ministry of Coal, Bernstein analysis

4. COMMERCIALS: 'COAL SYNGAS VS IMPORTED LNG'

COAL GASIFICATION VS. NATURAL GAS - UREA

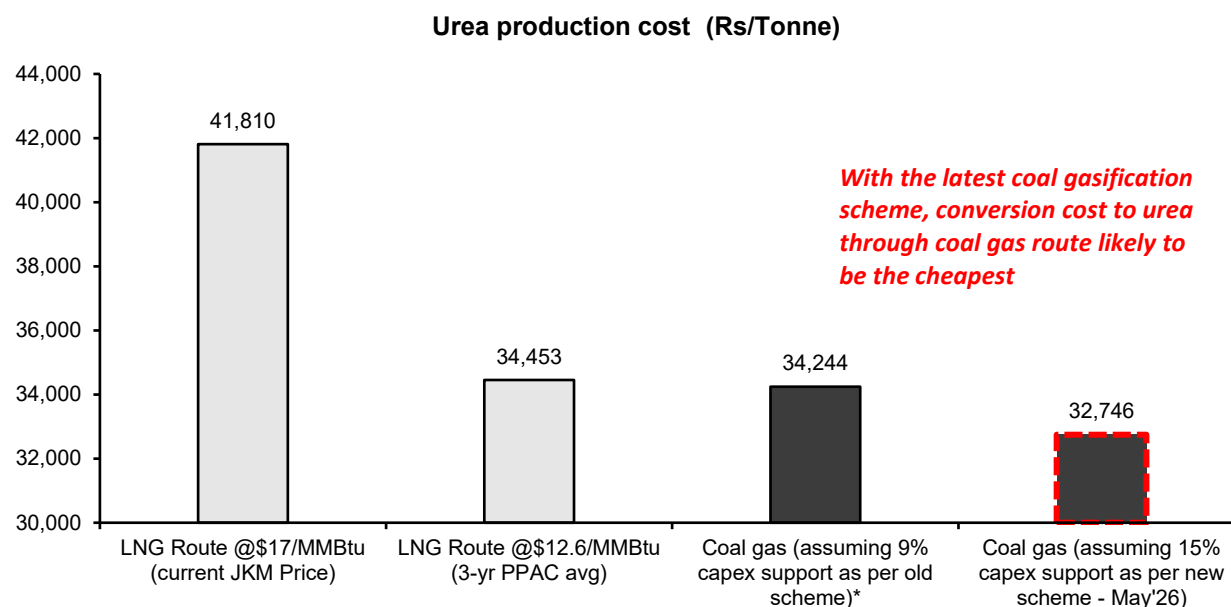
We could not find a lot of industry benchmarks for commercials of coal gasification. Hence, we rely on capex data from already announced plants in the country to compare coal gasification to LNG use case.

Coal gasification route to make Urea: We consider the Talcher plant capex (adjusting for some delays) to be the base case for capex, assume plant utilization at 80% (we don't have precedence for this, but given the low variable cost for gasification we assume one would sweat the plant to the maximum extent possible - but the high ash content of Indian coal could be a limitation), and we assume O&M cost at 2.25% of capex. Capex support - we show both the old scenario of just VGF support and new scenario of ~15% capex support.

LNG route to make Urea: For capex we assume Rs 62 Bn for a 1 MTPA Urea plant. For LNG price, we analyze both - 3-year average LNG price of \$12.6/MMBtu and the current spot JKM price of \$17/MMBtu (as of May'26), and consider a forex of 95 Rs/USD.

[Exhibit 17](#) shows the outcome of our analysis - coal and LNG routes are at parity at the 3 year avg LNG price, but coal gasification becomes more economical once if we compare to today's higher LNG price (\$17/MMBtu). Further if we assume the new subsidy, coal gasification is cheaper even against the 3 year avg. LNG price. Another important aspect we have to remember is that - the LNG route is heavily dependent on gas price and Forex rates, whereas the coal route is 100% indigenous and predictable - which is a very big strategic advantage. We also show a break-even line at different gas prices in [Exhibit 19](#).

EXHIBIT 17: Urea production cost via LNG vs. Coal Gasification



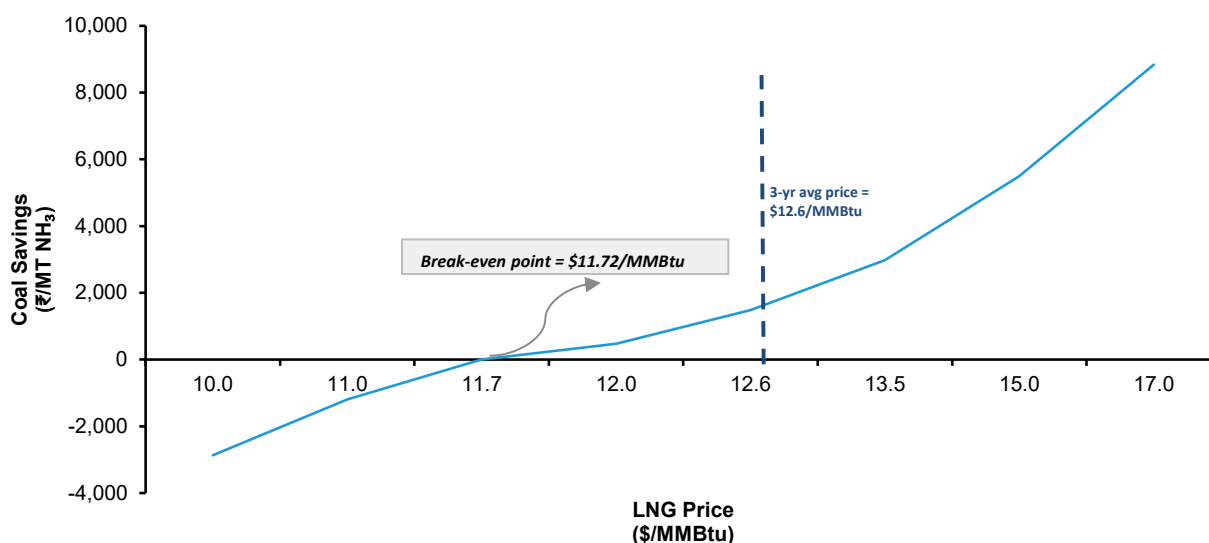
For old scheme- Assumed 9% capex support as ceiling of Rs.13.50 Bn was hit
Source: Petronac, PPAC, PIB, Bernstein estimates and analysis

EXHIBIT 18: **Cost breakdown for Urea production - LNG vs. Coal**

Cost Component	LNG Route (\$12.6/MMBtu)	Coal Gasification Route (Post-VGF Jan'24)	Notes
A. Feedstock	₹23,100 (68%)	₹8,500 (24%)	LNG: 17.6 MMBtu/MT × ₹1,315/MMBtu. Coal syngas: 17.6 MMBtu/MT × ₹484/MMBtu (all-in gate cost)
B. Annualised CAPEX	₹8,000 (23%)	₹21,200 (60%)	LNG: ₹6,500 Cr plant; Coal: ₹17,000 Cr plant. CRF @ 11% WACC, 20-yr tenor
C. O&M	₹1,400 (4%)	₹3,800 (10%)	3.5% of CAPEX per annum ÷ 720,000 MT capacity of TFL plant; 80% capacity utilization
D. Utilities	₹1,800 (5%)	₹2,500 (6%)	Coal route requires Air Separation Unit (~200 MW) vs. simpler SMR for LNG
E. VGF Benefit	—	₹ -1,600	GoI Viability Gap Funding for eligible PSU JV only; ₹1,350 Cr × CRF ÷ 720,000 MT
Total Cost (₹/MT NH ₃)	₹ 34,300	₹ 34,400	(rounded off to the nearest '000)

All numbers are rounded-off to nearest '00s

Source: Ministry of Coal, Bernstein analysis & estimates

EXHIBIT 19: **Breakeven for Urea - We see a breakeven at ~\$11.72 /MMBtu of gas, which has become lower than the 3-yr avg price with the recent capex subsidy****Sensitivity Analysis - Urea**

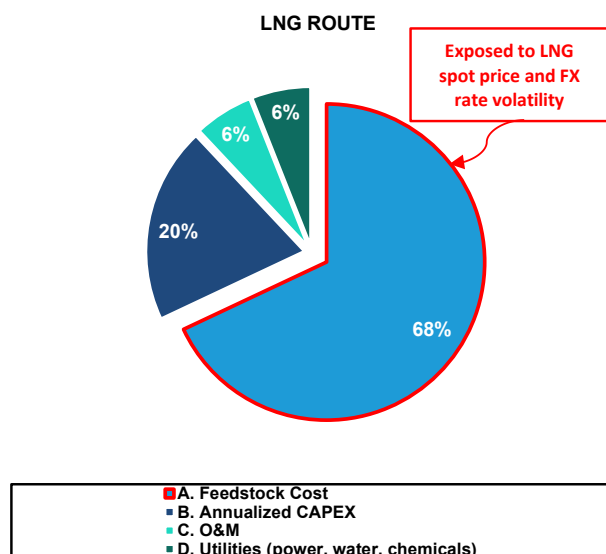
Source: Bernstein analysis and estimates

THE COST STRUCTURE BREAKDOWN

[Exhibit 20](#) and [Exhibit 21](#) present the cost component breakdowns for LNG and coal routes respectively. **LNG Route cost structure:** 68% feedstock — exposed to LNG spot price and FX volatility. The remaining 32% (CAPEX 20%, O&M 6%, utilities 6%) is largely rupee-denominated and fixed. This means 63% of the cost is unpredictable, and structurally depreciating in rupee terms every year.

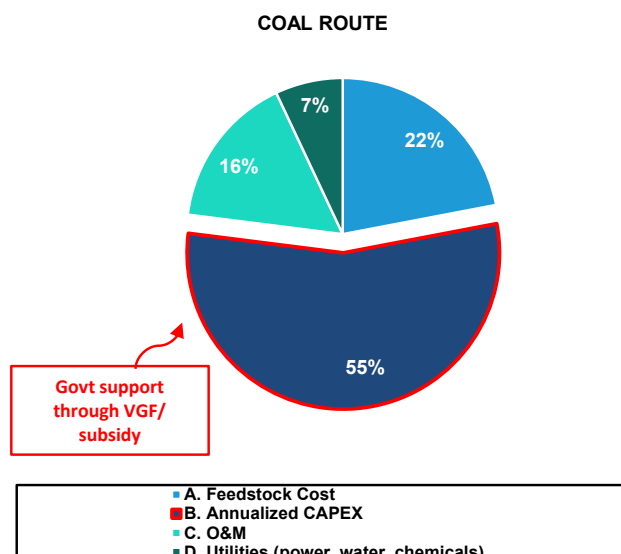
Coal Route cost structure: 20% feedstock, 59% CAPEX (front-loaded, fixed), 15% O&M, 6% utilities. VGF grant and DoF IRR guarantee exist specifically to address this CAPEX-heavy structure. Every year of rupee depreciation reduces the LNG route's feedstock cost advantage — which is why the economics are converging without any change in the coal route's absolute cost.

EXHIBIT 20: Cost Component Breakdown - LNG



Source: Bernstein analysis & estimates

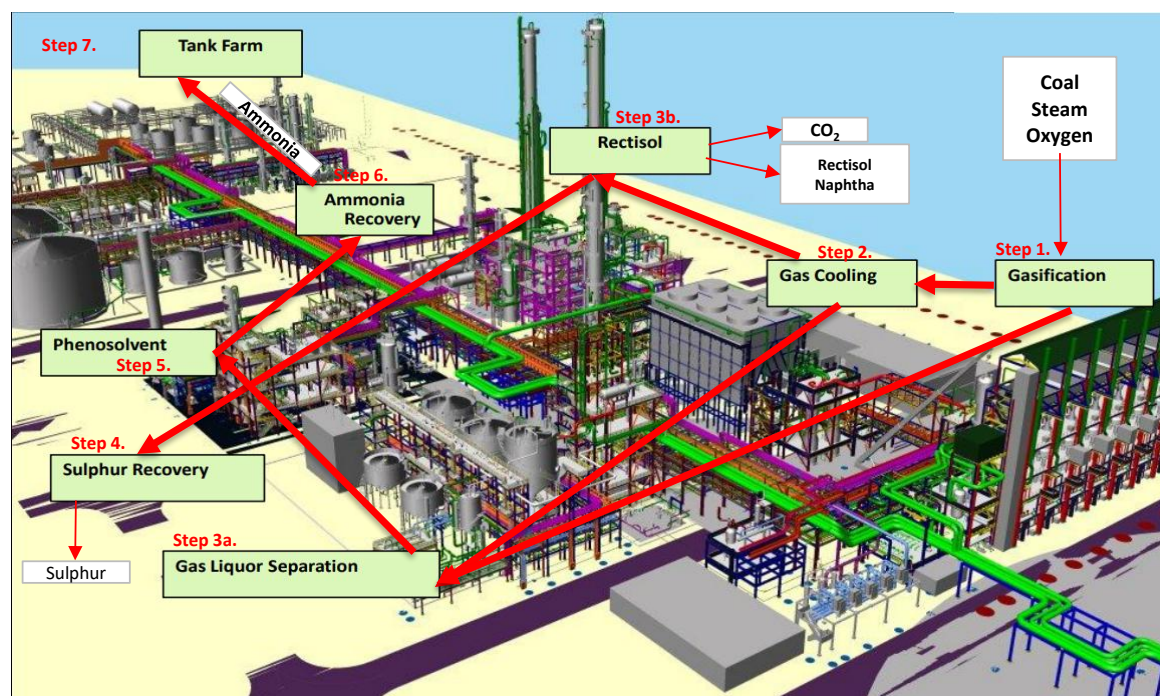
EXHIBIT 21: Cost Component Breakdown - Coal



Source: Bernstein analysis & estimates

Below we have shown the layout of JSPL's Angul gasification plant for reference.

EXHIBIT 22: JSPL's plant in Angul, Odisha



Source: JSPL archives, Bernstein analysis

KEY RISKS

1. Absence of Long-Term Offtake Agreements Beyond TFL (where DoF is the de facto off-taker through subsidy price support), no formal long-term offtake contracts exist for coal gasification output from BCGCL, CGIL, or private projects. SNG from CGIL enters the GAIL grid on a tariff arrangement — but tariff revision risk after the initial period is unquantified. Ammonium nitrate from BCGCL has no guaranteed buyer. *Government response in progress:* The Ministry of Coal is developing

a "Syngas Pricing Policy" that would mandate GAIL to offtake SNG at a formulaic price tied to imported LNG, providing implicit long-term offtake support. For fertiliser projects, DoF's existing Urea New Investment Policy (NIP) provides a production-linked subsidy that effectively backstops demand — but NIP coverage for coal gasification units requires explicit extension, currently under review.

2. Return Profile Uncertainty for Private Sector Projects The DoF 12% IRR guarantee applies only to TFL. BCGCL and CGIL have no equivalent government return backstop — their economics depend entirely on product prices (ammonium nitrate and SNG respectively) that are subject to market fluctuation. *Government response in progress:* The Ministry of Coal's proposed "Coal Gasification Development Fund" (budget allocation sought in FY2027 Union Budget) would provide a price support mechanism analogous to the CERC's pooled tariff for renewable energy — setting a minimum syngas purchase price that guarantees an 11–12% project IRR regardless of LNG market moves.

3. Technology Risk - Handling high ash Indian coal reduces the efficiency of the plant and leads to energy conversion losses as well as an accelerated wear and tear of the gasifiers. *Government response in progress:* MoC explicitly permits petcoke blending at regulated ROM coal prices. BHEL is indigenously developing fluidized bed gasifiers which will be able to handle high-ash, low-carbon coal. Research institutes like CSIR-CIMFR are also working to optimize high-ash coal utilization by Indian gasification plants.

4. Competition from renewable energy sources - Green ammonia is a cleaner optional with improving commercials, which could limit the share of coal gasification going forward if it becomes more attractive.

5. KEY PLAYERS

State owned companies dominate the market: Most of them playing dual roles. *i) BHEL* is - a) Leading the technology initiative to process Indian coal, b) Is also an equity holder in the BCGCL project. *ii) Coal India* is a) Fuel supplier, b) Equity shareholder in all 3 JVs - TFL, BCGCL and CGIL. *iii) GAIL* a) participates through TFL and CGIL equity, b) is also an infrastructure provider. Further, NTPC has a target to produce 5-10 MTPA of syngas by 2028-29, and L&T recently won an order to construct Coal-to-Ammonia Nitrate project in Odisha. Beyond them, among asset owners - JSPL (not covered) stands apart as the only company successfully operating its coal gasification plant at Angul. List of potential beneficiaries is given in [Exhibit 23](#).

EXHIBIT 23: **Key beneficiaries of India's coal gasification mission**

Company	Comments
<i>A — Coal / lignite feedstock + project developers</i>	
Coal India	Partner in multiple JVs for coal gasification projects (with GAIL, BHEL, BPCL, etc.)
NTPC	Target to produce 5-10 MTPA of syngas by 2028-29
NLC India	NLC India's Rs 44Bn lignite-to-methanol project likely to be completed by Mar'27
GMDC	GMDC inks MoU with NTPC to explore coal, lignite gasification opportunities
Adani Enterprises	India's largest private commercial coal miner
<i>B — Gasifier technology, equipment, R&D & EPC</i>	
BHEL	JV with Coal India for coal gasification project; Working on coal-to-methanol demo plants
Thermax	Operating coal-to-methanol pilot plant with IIT Delhi at Pune
L&T	Recently won a large order BCGCL for a Coal-to-Ammonia-Nitrate project in Odisha
Engineers India	Signed MoU with NTPC in Oct 2025 to develop coal-to-SNG facility
<i>C — JV equity partners & downstream producers (SNG, fertilizer, chemicals)</i>	
Gail	49% equity in CIL-GAIL JV in SNG plant, WB (project cost- Rs 131 Bn)
Rashtriya Chemicals	31.85% equity in Talcher Fertilizers which is building a coal gasification based 1.27 MTPA urea plant
<i>D — Private sector with operational gasification / announced projects</i>	
Jindal Steel & Power	Operates India's only coal-based syngas DRI plant at Angul, Odisha

We cover NTPC and L&T only

Source: Company filings, Press articles, Bernstein analysis

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VALUATION METHODOLOGY**Larsen & Toubro Ltd**

We set a target price of Rs 4,637 using a SOTP approach. We consider their subsidiaries like LTI Mindtree, L&T technology services and L&T finance holdings with adjustments (like holding company discount) to market cap and value non-services business using forward P/E ratio of 30x on our FY28 EPS estimates.

NTPC Ltd

We value NTPC using DCF method to arrive at our INR430 price target. We forecast the financials till 2070 asset wise - assuming no further thermal/nuclear capacity addition. For renewables, we account for capacity addition till FY40, post which we separately value the renewable business assuming a 12% market share for capacity additions till FY60. We have used a weighted average cost of capital (WACC) of 8.0%. We also allocate a 30% holdco discount to NTPC Green's valuation.

Power Finance Corp Ltd

We value PFC using SOTP basis - a) PFC standalone: FY27 earnings, give a multiple of 7x to standalone EPS of PFC (Ex of REC dividend); b) REC target value with 25% holding company discount. Our target price is Rs. 460 which implies a P/B of 1.1x on FY26E book (ex of non-controlling interests)

Rural Electrification Corporation Ltd

We value REC at 7x multiple to its FY27 earnings forecast. Our price target is Rs. 450 which implies a P/B of 1.3x on FY26E book.

RISKS**Larsen & Toubro Ltd**

Downsides:

- 1) Prolongation of the M.East conflict
- 2) Shift in existing government stance towards infrastructure investments
- 3) Rise in commodity prices - steel, cement etc.
- 4) Rising rates impacting capex cycle
- 5) Challenges in executing projects in Middle-East due to local sourcing requirements or working capital stress
- 6) Slowdown in Middle East capex

NTPC Ltd

Downside:

1. Power demand in India growing slower than our anticipation of 5-6% CAGR
2. Rapid reduction in battery prices reducing the relevance of adding a lot thermal capacity
3. Further disappointment in renewable capacity addition by its subsidiary
4. Nuclear - No traction in nuclear project award

Power Finance Corp Ltd

Downside risks:

1. Rise in exposure to non-PPA private loans- Merchant renewables, pumped-storage or Green H2 without off-take guarantee. Rise in such exposures could lead to re-occurrence of stressed assets.
2. DISCOM bail-outs: Historically bailouts have prevented NPAs but might result in rate reduction. Another round of bailout could lead to lower ROE for PFC and loan book shrinking
3. Rise in private infra lending which is not their core expertise
4. Competition from banks once their liquidity improves- could lead to lower spreads

Rural Electrification Corporation Ltd

Downside risks:

1. Rise in exposure to non-PPA private loans- Merchant renewables, pumped-storage or Green H2 without off-take guarantee. Rise in such exposures could lead to re-occurrence of stressed assets.
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3. Rise in private infra lending which is not their core expertise
4. Competition from banks once their liquidity improves- could lead to lower spreads

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- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

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The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 500 Banks And Financial Services Index (BEBANKS) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 500 Insurance Index (BEINSUR) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

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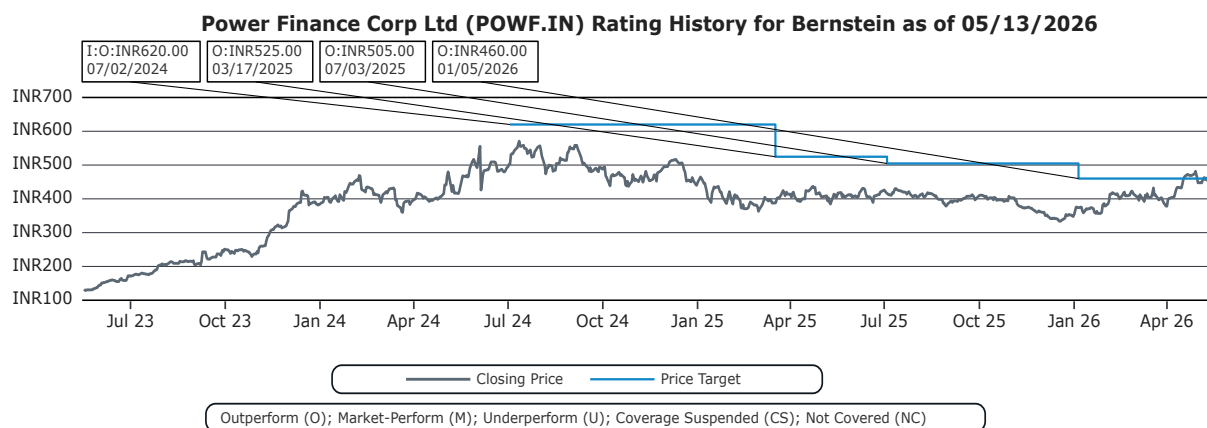
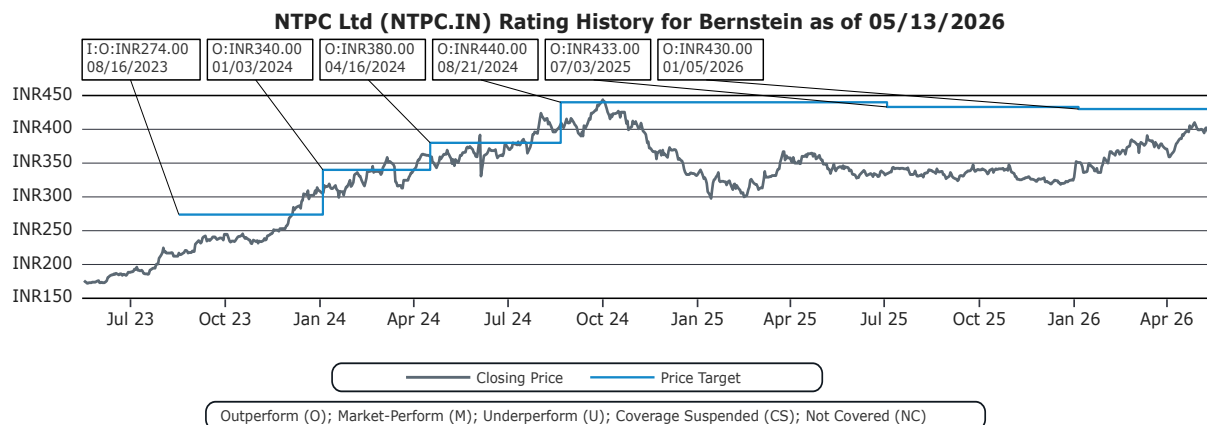
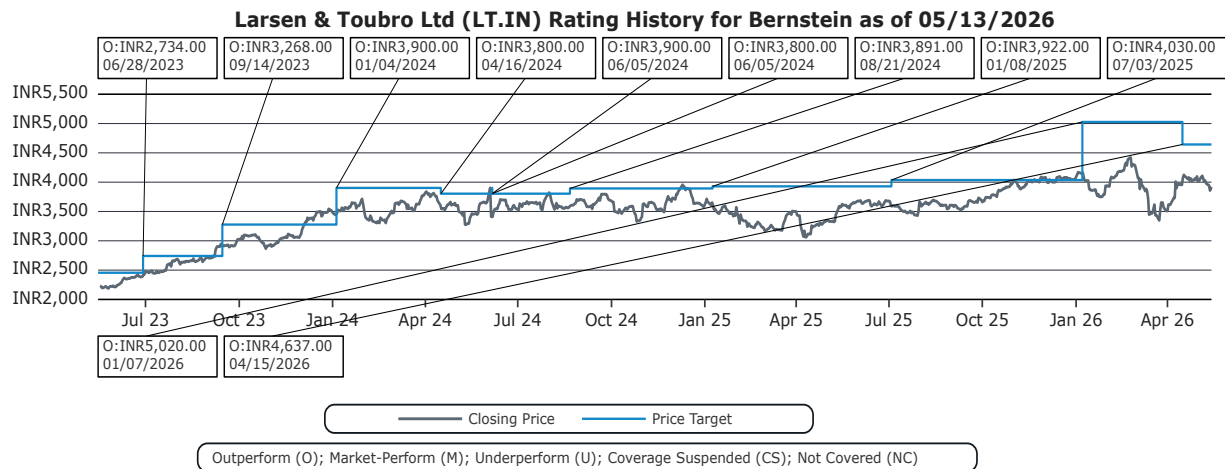
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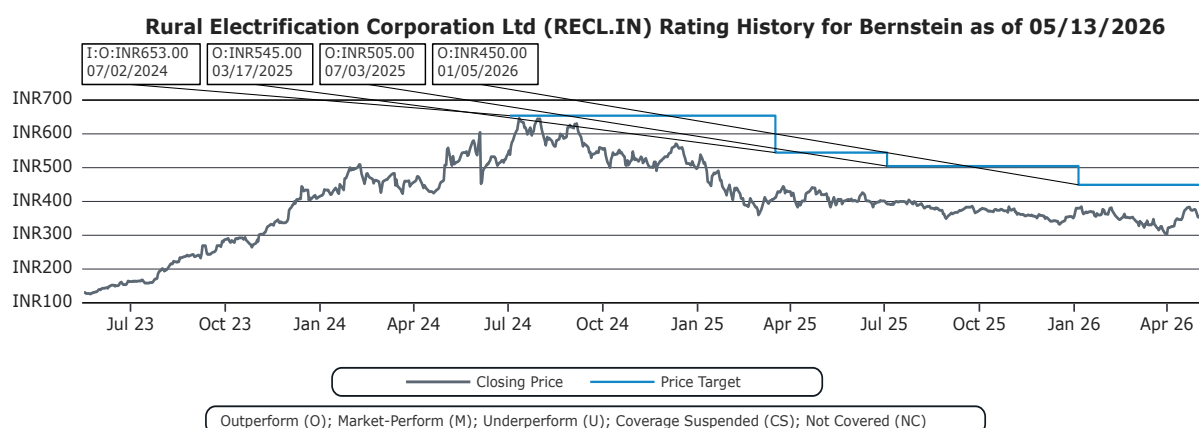
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